

1 Case Study 2: Institutional Benchmarking

1.1 Objective

Compare publication output, citation impact, and research profiles of five universities using size-independent and field-normalised indicators.

1.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

1.3 Data acquisition

```
institutions <- tribble(
  ~short_name,    ~oa_id,
  "MIT",          "I63966007",
  "Oxford",       "I40120149",
  "ETH Zurich",   "I202697423",
  "U São Paulo",  "I66491032",
  "U Cape Town",  "I173603587"
)

fetch_works <- function(inst_id) {
  res <- tryCatch(
    oa_fetch(
      entity = "works",
      authorships.institutions.id = inst_id,
      from_publication_date = "2020-01-01",
      to_publication_date = "2022-12-31",
      type = "article",
      options = list(sample = 300, seed = 42)
    ),
    error = function(e) NULL
  )
  if (is.null(res)) tibble() else res
}

inst_data <- institutions |>
  mutate(works = map(oa_id, fetch_works)) |>
  filter(map_int(works, nrow) > 0)
```

1.4 Publication volume

```
inst_data |>
  mutate(n = map_int(works, nrow)) |>
  ggplot(aes(x = n, y = reorder(short_name, n))) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Sampled articles", y = NULL) +
  theme_sci()
```

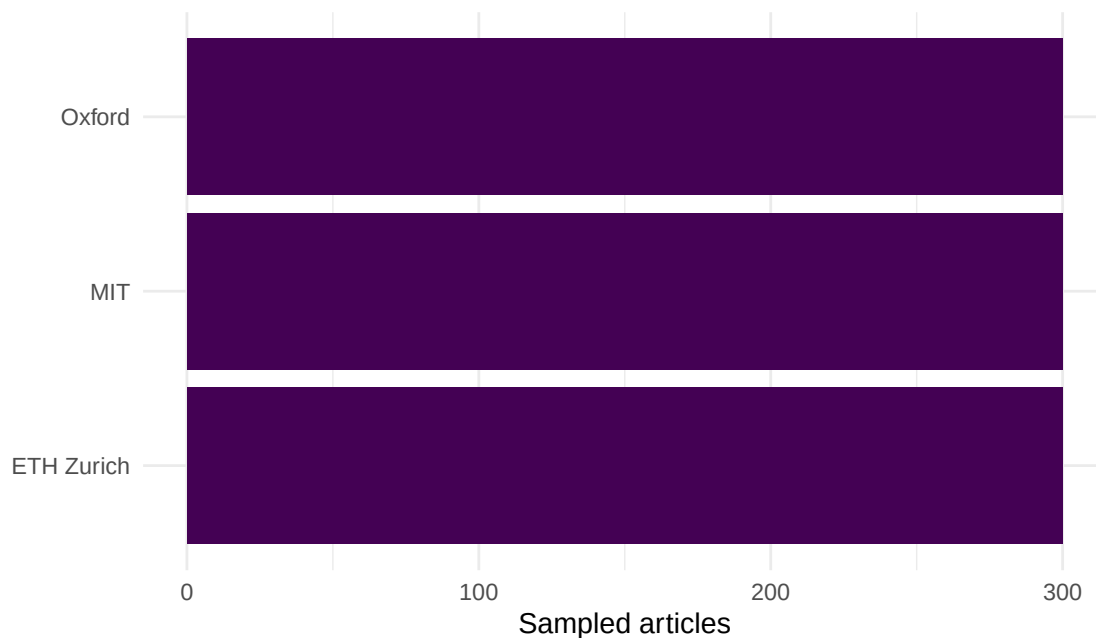


Figure 1: Sampled publication counts by institution.

1.5 Citation impact

```
impact <- inst_data |>
  mutate(
    mean_cites = map_dbl(works, \(w) mean(w$cited_by_count, na.rm = TRUE)),
    median_cites = map_dbl(works, \(w) median(w$cited_by_count, na.rm = TRUE)),
    h_index = map_int(works, \(w) compute_h_index(w$cited_by_count))
  ) |>
  select(short_name, mean_cites, median_cites, h_index) |>
  arrange(desc(mean_cites))

impact |>
  gt() |>
  fmt_number(columns = mean_cites, decimals = 1)
```

short_name	mean_cites	median_cites	h_index
MIT	41.0	16.5	55
Oxford	39.0	15.0	54
ETH Zurich	39.0	11.5	48

short_name	full_count	frac_count	ratio
MIT	300	115	0.38
Oxford	300	103	0.34
ETH Zurich	300	110	0.37

1.6 Full vs. fractional counting

```
counting <- inst_data |>
  mutate(
    full_count = map_int(works, nrow),
    frac_count = map_dbl(works, function(w) {
      w |>
        select(id, authorships) |>
        unnest(authorships, names_sep = "_") |>
        unnest(authorships_affiliations, names_sep = "_") |>
        group_by(id) |>
        summarise(n_inst = n_distinct(authorships_affiliations_id),
                  .groups = "drop") |>
        mutate(frac = 1 / pmax(n_inst, 1)) |>
        pull(frac) |>
        sum()
    })
  ) |>
  select(short_name, full_count, frac_count) |>
  mutate(ratio = round(frac_count / full_count, 2))

counting |> gt()
```

```
counting |>
  pivot_longer(c(full_count, frac_count), names_to = "method", values_to = "count") |>
  mutate(method = recode(method, full_count = "Full", frac_count = "Fractional")) |>
  ggplot(aes(x = count, y = reorder(short_name, count), fill = method)) +
  geom_col(position = "dodge", width = 0.7) +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = "Count", y = NULL, fill = "Method") +
  theme_sci()
```

1.7 OA rates

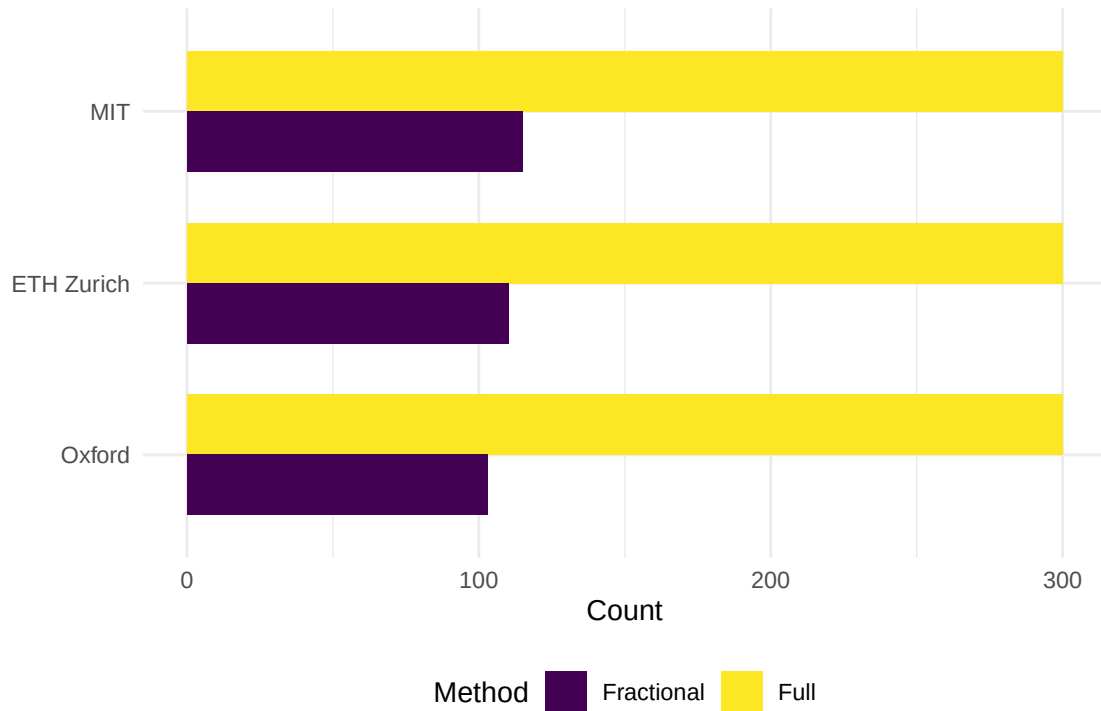


Figure 2: Full vs. fractional publication counts.

```
inst_data |>
  mutate(oa_rate = map_dbl(works, \(w) mean(w$oa_status != "closed", na.rm = TRUE))) |>
  ggplot(aes(x = oa_rate, y = reorder(short_name, oa_rate))) +
  geom_col(fill = palette_sci(1)) +
  scale_x_continuous(labels = scales::percent) +
  labs(x = "OA rate", y = NULL) +
  theme_sci()
```

1.8 Key findings

1. **Size vs. impact:** Raw publication counts favour larger institutions. Fractional counting and h-index provide size-independent comparisons.
2. **Collaboration intensity:** Institutions with lower fractional/full ratios have more inter-institutional collaborations.
3. **OA variation:** OA rates differ substantially, reflecting different national mandates and institutional policies.
4. **No single ranking:** Different indicators produce different orderings. This illustrates the Leiden Manifesto principle that multiple indicators are needed for fair comparison [hicks2015].

1.9 Lessons learned

- Sampling introduces uncertainty: these are illustrative comparisons, not definitive rankings.
- Field mix confounds all comparisons: a medical university and a technical university are not directly comparable without field normalisation.

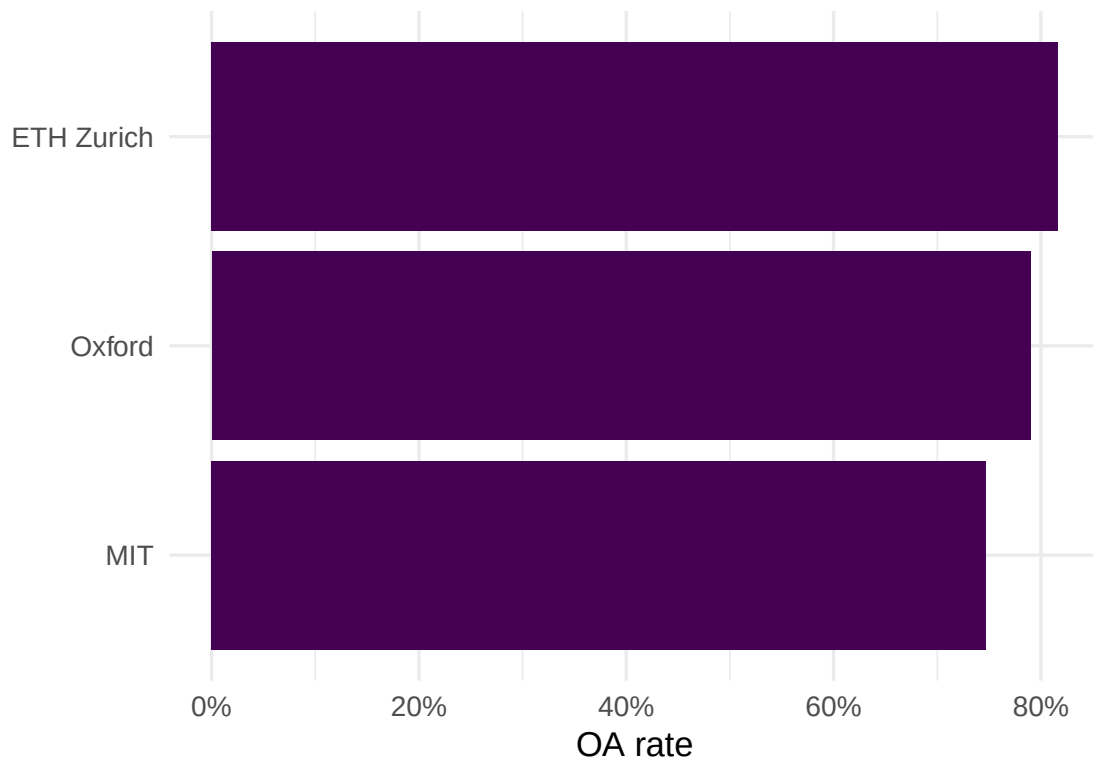


Figure 3: Open access rate by institution.

- Responsible benchmarking requires reporting the counting method, normalisation approach, and sample size transparently.